HOLY CROSS COLLEGE

Life to the Full

11 Physics

Nuclear Physics

ATOMS, ISOTOPES AND RADIOISOTOPES

Atoms of elements are composed of protons and neutrons (referred to as ***nucleons***) in a nucleus (nuclei are also referred to as ***nuclides***) with electrons in different energy level orbits around the nucleus. The nucleus contains nearly all of the mass of the atom, but accounts for less than a million millionth (10^{-12}) of its volume.

The atomic number of an element is equal to the number of protons in each of its atoms. It is this that determines which element an atom represents. The atomic number is also equal to the number of electrons in orbit around the atom.

Atomic number (Z) = number of protons = number of electrons in a neutral atom.

The mass number of an element is the sum of the numbers of protons and neutrons in the nucleus.

Mass number (A) = number of protons + number of neutrons in the nucleus of an atom.

Atoms are represented by the symbol ${}^{A}_{Z}X$.

e.g. ${}^{235}_{92}U$ represents Uranium - 235.

The atomic mass of an element (often incorrectly referred to as its mass number) is the average mass of its atoms. Atomic mass takes into account that not all atoms of an element have the same mass due to the presence of isotopes of that element.

Atomic mass = average mass of the nuclides of an element (isotopes considered).

Isotopes are atoms of elements that have ***different numbers of neutrons*** in their nuclei. Representing the same element, these atoms must have identical numbers of protons in those nuclei. Isotopes have the ***same chemical properties*** (they react and bond with other atoms in precisely the same way) but ***different physical properties***.

Radioisotopes are naturally occurring isotopes that are unstable. An unstable nucleus may spontaneously lose energy by emitting a particle and so change into a different element or isotope. Unstable atoms are radioactive and an individual radioactive isotope is known as a radioisotope.

e.g. Carbon has two stable isotopes (C-12 and C-13) and one isotope in nature that is not stable (C-14).

The nucleus of a radioactive carbon-14 atom may spontaneously decay, emitting high-energy particles that can be dangerous.

Physics file

The heaviest stable isotope in the universe is ${}^{209}_{83}Bi$. Every isotope of every element with more than 83 protons, i.e. beyond bismuth in the periodic table, is radioactive. For example, every isotope of uranium (Z = 92) is radioactive. Technetium (Z = 43) and promethium (Z = 61) are the only elements with an atomic number below bismuth (Z = 83) that do not have any stable isotopes. Uranium is the heaviest element that occurs naturally on Earth. All of the elements with atomic numbers greater than 92 have been artificially manufactured.

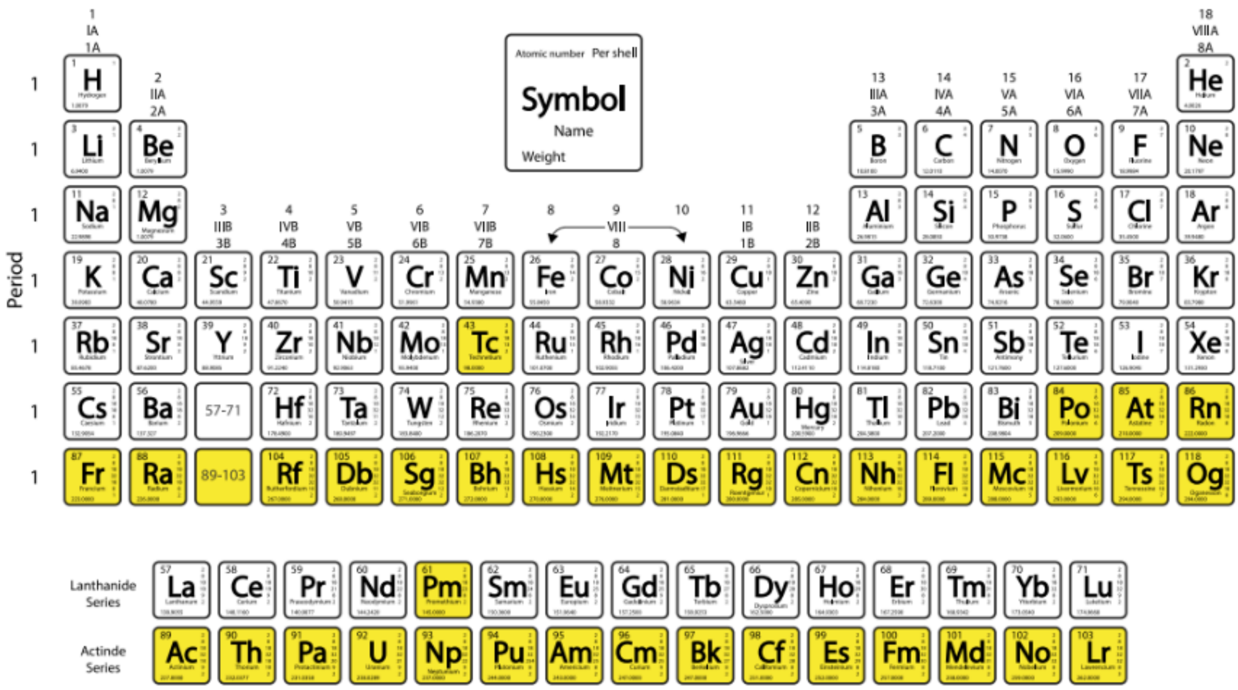


FIGURE 3.1.7 The periodic table of the elements. All isotopes of the elements shaded in yellow are radioactive.

Extra Information

Protons, neutrons and electrons are ***not*** the smallest sub-atomic particles.

Physics in action — Quarks!

Our understanding of the atom has changed greatly in the past 100 years. It was once thought that atoms were like miniature billiard balls: solid and indivisible. The word atom comes from the Greek 'atomos' meaning indivisible. However, the first subatomic particles—the electron, the proton and then the neutron—were discovered in the period 1897–1932.

Since World War II, further research has uncovered some 200 other subatomic particles.

Table 5.1 Fundamental particles

Quarks	Leptons
Up	Electron
Down	Muon
Charm	Neutrino
Strange	Tau
Top	Electron-neutrino
Bottom	Tau-neutrino

Examples of these include pi-mesons, mu-mesons, kaons, tau leptons and neutrinos. For many years, physicists found it difficult to make sense of this array of subatomic particles. Then in 1964 Murray Gell-Mann put forward a simple theory. He suggested that most subatomic particles were themselves composed of a number of more fundamental particles called quarks. For example, the proton and the neutron would be made of three quarks while all the mesons would be made of two quarks.

A significant amount of effort has been directed to testing his theory—both theoretically and experimentally—and today the quark theory is accepted. Currently, it is accepted that there are six different quarks, each with different properties. As seen in Table 5.1, quarks have unusual names!

Subatomic particles that consist of quarks are known as hadrons. Hadrons are the more massive subatomic particles. The proton consists of two 'up' quarks and one 'down' quark, while neutrons consist of one 'up' quark and two 'down' quarks.

Apart from the hadrons, there are the light particles—the leptons. These are indivisible, point-particles like the electron. Leptons are not composed of quarks.

The theory suggests that quarks and leptons are the ultimate fundamental particles. They cannot be further divided.

TYPES OF RADIATION

Why Radioactive Nuclei are Unstable

Radioactivity is the name given to the radiations emitted by certain elements that are unstable. It was determined by experiment that radioactivity was due to the disintegration or decay of the nucleus by the emission of a particle or particles, and / or energy in the form of light.

The nucleus, particularly that of a large atom, is unstable and contains too much energy. In order to become more stable, it emits either a particle (that carries energy away as E_k) or light in the form of gamma rays (very high energy light).

Inside the nucleus, there are two completely different forces acting.

- ***An electric force of repulsion between the protons***. On its own, this would blow the nucleus apart, so clearly a second force must act to bind the nucleus together.

- ***The nuclear force***, a powerful force of attraction between nucleons, which acts only over a very short range.

In a stable nucleus, there is a delicate balance between the repulsive electric force and the attractive strong force.

There appears to be a connection to the presence of neutrons. ***The nuclear force is found to weaken when too few or too many neutrons are present***.

i.e. The diagram at right shows the relationship between proton and neutron number for the elements.

For Z < 40, number of protons and neutrons are about the same. These are very stable.

For larger nuclei, the neutron number becomes bigger than the proton number. This reduces the repelling force between the protons and makes the nucleus stable.

For Z > 83, the nuclear force is not strong enough to overcome the repulsive electrostatic force between the protons. All of these nuclei are radioactive.

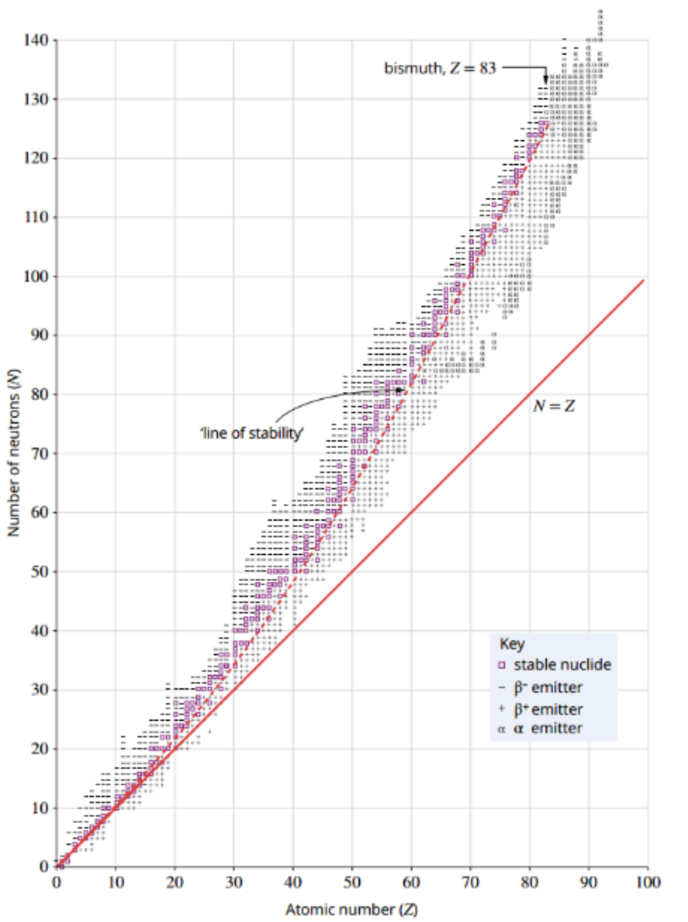


FIGURE 3.1.10 A chart showing stable and radioactive isotopes, plotted according to their number of protons (atomic number) and number of neutrons.

History of Radiation

The accidental discovery of radioactivity by Henri Becquerel in 1896 began a quest for understanding that has had far-reaching effects on life on Earth. The existence of radioactive particles caused scientists to wonder where they came from. The realisation that the nucleus of an atom was not a solid, unbreakable entity, but consisted of other particles that could be released in extreme conditions was emerging.

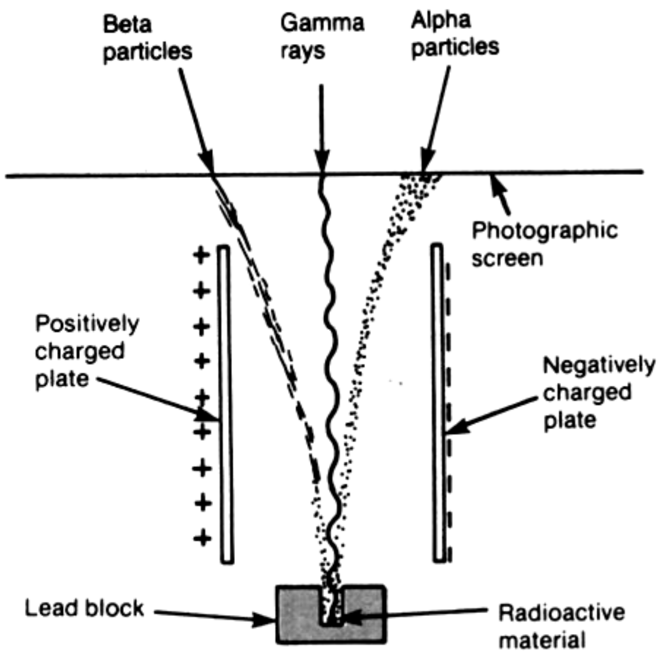
- 1896 Henri Becquerel discovered radioactivity, accidentally observing that a uranium salt emitted rays that darkened a photographic plate.
- 1898 Rutherford showed that Becquerel's radiation consisted of alpha particles and beta particles.
- 1900 Paul Villard discovered gamma rays , the third component of Becquerel 's radiation.
- 1902 Rutherford and Frederick Soddy proposed that the emission of radioactivity was the result of radioactive transformations. These occur when a nucleus splits into two smaller nuclei. It emits radioactivity.

Types of Radiation

Ernest Rutherford and Paul Villard discovered that there were three different types of emission from radioactive substances. They named these ***alpha (α)***, ***beta (β)*** and ***gamma (γ)*** radiation.

Experiments showed that the alpha and beta emissions were actually particles expelled from the nucleus. Gamma radiation was found to be high-energy electromagnetic radiation, also emanating from the nucleus.

These radiations were identified by passing them through a set of parallel plates with a large potential difference (voltage) across them.



This experiment showed the following about the various radiations.

Alpha (α) particle
- Positively charged (attracted to negative plate).
- Relatively large mass (takes longer to "bend" in its path).

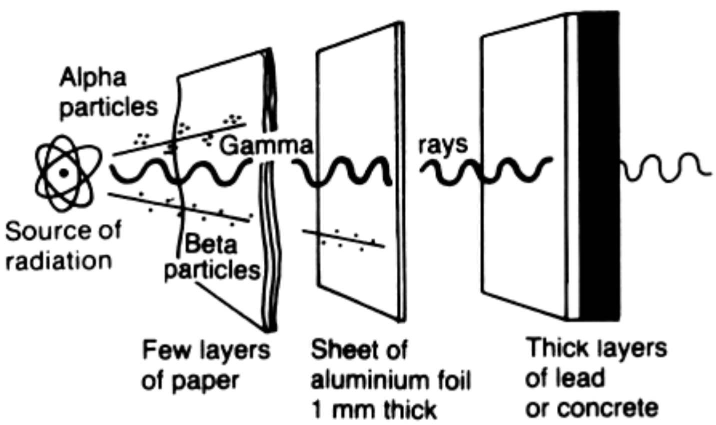
Beta (β) particle
- Negatively charged.
- Relatively light particle (its path "bends" quickly).

Gamma (γ) Ray
- No charge (not deflected).
- No mass.
- Pure energy in the form of electromagnetic radiation (light).

Properties Of Radiation

	α	β	γ
Symbol	$({}^{4}_{2}He)^{2+}$	${}^{0}_{-1}e$	${}^{0}_{0}\gamma$
Charge	+ 2	- 1	0
Mass	Approx. 4u	9.11 x 10^{-31} kg	0
Speed	~ 0.1 c	~ 0.9 c	c
Penetration	Stopped by a few cm of air. (Big and slow means lots of collisions.)	Stopped by a few metres of air, 1-2 mm of Al.	Mostly absorbed by a few cm of Pb - never completely absorbed.
Ionising Effect	Very high. (Large size makes it easy to remove electrons from atoms.)	Not great due to its small size.	Very small since it is pure energy and not a particle.

- 1u = 1.66 x 10^{-27} kg
- c = speed of light = 3.00 x 10^{8} ms^{-1}



TRANSMUTATION OF RADIOACTIVE ELEMENTS

Transmutation: the change of one element into another by the emission of radiation.

Radioactive substances disintegrate spontaneously, emitting α or β particles (usually accompanied by γ rays). After emitting an α or β particle, the nucleus is still unstable and the emission of a γ ray removes the excess energy and the nucleus returns to a stable state.

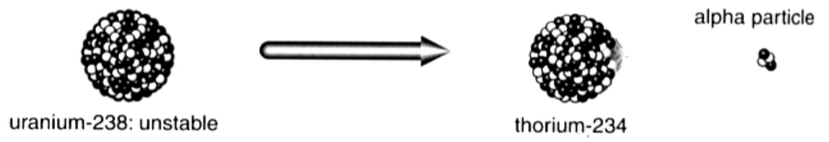
The process of transmutation is represented by ***nuclear equations***. The rules for these equations are as follows.

- The algebraic sum of the atomic numbers of the reacting species equal the algebraic sum of the product species. ***This conserves charge***.

- The algebraic sum of the mass numbers of the reacting species equal the algebraic sum of the product species.

Alpha Decay

An alpha particle is identical to a helium nucleus with no electrons. Hence its charge is +2. The symbol can be written as $({}^{4}_{2}\text{He})^{2+}$ or $({}^{4}_{2}\alpha)$.



$$ {}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\alpha + \text{energy} $$

The energy released is carried as E_k by the α particle.

Beta Decay

Beta particles are electrons, but they are electrons that have originated from the nucleus of a radioactive atom, not from the electron cloud. The symbol can be written as ${}^{0}_{-1}\text{e}$ or ${}^{0}_{-1}\beta$. The charge is -1 and the zero indicates its mass is far less than either a proton or neutron.

Physics file

A different form of beta decay occurs in atoms that have too many protons. An example of this is the radioactive decay of unstable nitrogen-12. There are seven protons and five neutrons in the nucleus, and a proton may spontaneously change into a neutron and emit a neutrino and a positively charged beta particle. This is known as a β^+ (beta-positive) decay and the positively charged beta particle is called a *positron*.

The equation for this decay is:

$$ {}^{12}_{7}\text{N} \rightarrow {}^{12}_{6}\text{C} + {}^{0}_{+1}\text{e} + \nu + \text{energy} $$

Positrons, ${}^{0}_{+1}\text{e}$ or ${}^{0}_{+1}\beta$, have the same properties as electrons, but their electrical charge is positive rather than negative. Positrons are an example of *antimatter*.

Beta decay occurs in nuclei where there is an imbalance of neutrons to protons. Typically, if a light nucleus has too many neutrons to be stable, a neutron will spontaneously change into a proton, and an electron and an uncharged massless particle (called an ***antineutrino*** $\bar{\nu}$) are ejected to restore the nucleus to a more stable state.

$$\text{i.e.} \quad {}^{1}_{0}\text{n} \rightarrow {}^{1}_{1}\text{p} + {}^{0}_{-1}\text{e} + \nu$$

C-14 is unstable with too many neutrons and undergoes a beta decay.

$${}^{14}_{6}\text{C} \rightarrow {}^{14}_{7}\text{N} + {}^{0}_{-1}\beta + \nu + \text{energy}$$

Physics file

Neutrinos are particles with the lowest mass in nature, and they permeate the universe. Neutrinos have no charge and their mass has only recently been discovered to be about one-billionth that of a proton, i.e. about 10^{-36} kg. While you have been reading these sentences, billions of neutrinos have passed right through your body, and continued on to pass right through the Earth! Fortunately neutrinos interact with matter very rarely and so are completely harmless. It has been estimated that if neutrinos passed through a piece of lead 90 light-years thick, they would still have only a 50% chance of being absorbed!

Gamma Decay

Generally, after a radioisotope has emitted an alpha or beta particle, the daughter nucleus holds an excess of energy. The protons and neutrons in the daughter nucleus then rearrange slightly and off-load this excess energy by releasing gamma radiation (high-frequency electromagnetic radiation). Gamma rays—like all light—have no mass and are uncharged and so their symbol is ${}^{0}_{0}\gamma$.

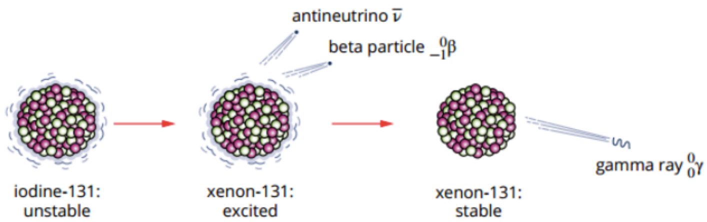


FIGURE 3.2.5 The gamma and beta decay of iodine-131.

$${}^{131}_{53}\text{I} \rightarrow {}^{131}_{54}\text{Xe} + {}^{0}_{-1}e + \gamma$$

Physics file

Gamma decay alone occurs when a nucleus is left in an energised or excited state following an alpha or beta decay. This excited state is known as a *metastable state* and it usually only lasts for a short time. An example of this is the radioactive decay of iodine-131, usually a two-stage process. First, a beta particle is emitted and the excited nuclide xenon-131m is formed. Then, the nucleus undergoes a second decay by emitting a gamma ray:

$${}^{131}_{53}\text{I} \rightarrow {}^{131m}_{54}\text{Xe} + {}^{0}_{-1}\text{e}$$

$${}^{131m}_{53}\text{Xe} \rightarrow {}^{131}_{53}\text{Xe} + \gamma$$

The 'm' denotes an unstable or metastable state. Cobalt-60 and technetium-99 also exist in metastable states.

Artificial Transmutation

Most of the radioisotopes that are used in industrial and medical applications are synthesised by artificial transmutation. There are now over 2000 such artificial radioisotopes.

e.g. One of the ways that artificial radioisotopes are manufactured is by ***neutron absorption***. (In Australia, this is done at the Lucas Heights reactor near Sydney.)

The neutron is absorbed into the nucleus and this creates an unstable isotope of the sample element, which usually becomes a beta emitter.

Cobalt-60 (widely used for cancer treatment) is manufactured from Cobalt-59.

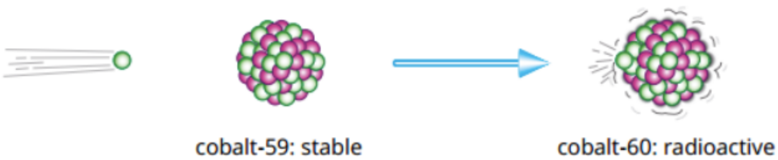


FIGURE 3.1.9 A neutron colliding with ^{59}Co nuclei to form ^{60}Co.

$${}^{59}_{27}\text{Co} + {}^{1}_{0}\text{n} \rightarrow {}^{60}_{27}\text{Co}$$

DETECTING RADIATION

Physics in action — How radiation is detected

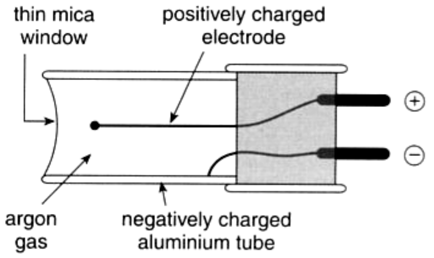


Figure 5.15

A radioactive emission that enters the tube in a Geiger counter will ionise the argon gas and cause a pulse of electrons to flow between the electrodes. This pulse registers as a count on a meter.

Figure 5.16

Film badges are used to monitor the exposure levels of doctors, radiologists, dentists and technicians who work with radiation.

Our bodies cannot detect alpha, beta or gamma radiation. A number of devices have therefore been developed to detect and measure radiation.

A common detector is the *Geiger counter*. These are used:

- by geologists searching for radioactive minerals such as uranium
- to monitor radiation levels in mines
- to measure the level of radiation after a nuclear accident such as the one at Chernobyl
- to check the safety of nuclear reactors
- to monitor radiation levels in hospitals and factories.

A Geiger counter consists of a Geiger–Müller tube filled with argon gas as shown in Figure 5.15. A voltage of about 400 V is maintained between the positively charged central electrode and the negatively charged aluminium tube. When radiation enters the tube through the thin mica window, the argon gas becomes ionised and releases electrons. These electrons are attracted towards the central electrode and ionise more argon atoms along the way. For an instant, the gas between the electrodes becomes ionised enough to conduct a pulse of current between the electrodes. This pulse is registered as a count. The counter is often connected to a small loudspeaker so that the count is heard as a 'click'.

People who work where there is a risk of continuing low-level exposure to radiation usually pin a small radiation-monitoring device to their clothing. This could be either a film badge or a TLD (thermoluminescent dosimeter). These devices are used by personnel in nuclear power plants, hospitals, airports, dental laboratories and uranium mines to check their daily exposure to radiation. When astronauts go on space missions, they wear monitoring badges to check their exposure to damaging cosmic rays.

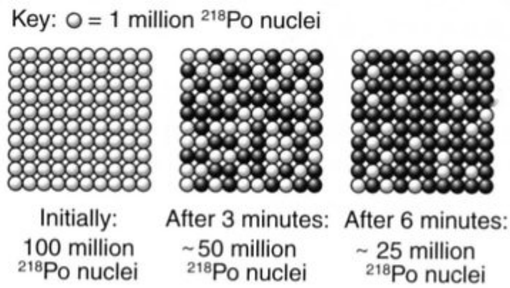
Film badges contain photographic film in a light-proof holder. The holder contains several filters of varying thickness and materials covering a piece of film. After being worn for a few weeks, the film is developed. Analysis of the film enables the type and amount of radiation to which the person has been exposed to be determined.

Thermoluminescent dosimeters are more commonly used than film badges. These contain a disk of lithium fluoride encased in plastic. Lithium fluoride can detect beta and gamma radiation as well as X-rays and neutrons. Thermoluminescent dosimeters are a cheap and reliable method for measuring radiation doses.

HALF-LIFE OF RADIOISOTOPES

Different radioactive elements decay at different rates. The rate of decay is directly proportional to the number of undecayed radioactive atoms remaining.

Consider the decay of a sample of Polonium-218, assuming 100 million atoms exist initially.

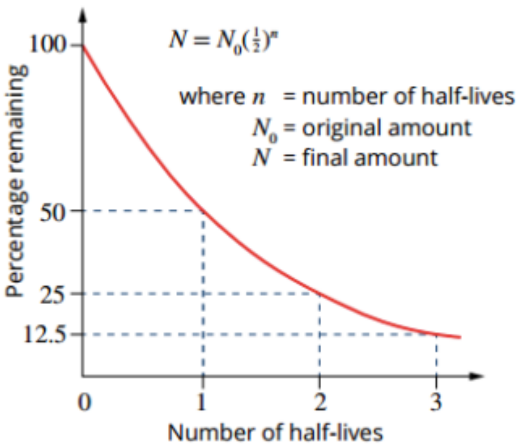


Every 3 minutes, half of the radioactive nuclei decay.

i.e.   The amount left decreases by half every 3 minutes.
       Hence the half-life is 3 minutes.

> ***Half-Life***: the time taken for half of the radioactive atoms in a sample to decay.

This relationship can be seen in the following diagram.



Time (t)	0	$t_{½}$	$2t_{½}$	$3t_{½}$	$4t_{½}$	$5t_{½}$	$nt_{½}$
Sample remaining	N_0	$\frac{N_0}{2}$	$\frac{N_0}{4}$	$\frac{N_0}{8}$	$\frac{N_0}{16}$	$\frac{N_0}{32}$	$\frac{N_0}{2^n}$

From the graph **it can be seen that** the decay reduces exponentially in the following way.

- After **two half-life periods,** only 1/4 of the radioactive atoms remain.
- After **three half-life periods, only 1/8** of the radioactive atoms remain.
- After **four half-life periods, only y 1/16** of the radioactive atoms remains.

Thus the fraction of radioactive atoms remaining after a given time can be calculated by:

$$\boxed{\text{Fraction left} = \frac{1}{2^n}}$$

where   n = the number of half-life periods of time.

This fraction can be used to determine the number of undecayed atoms left in a sample, the mass of the radioactive element left or the radiation count given off by the remaining undecayed atoms.

The decay equation can then be written as:

$$\boxed{N = N_0 \frac{1}{2_n}}$$

where N = number of undecayed atoms, mass of remaining radioactive atoms or" the radiation count given off by the undecayed atoms.

N_0 = the original number of atoms, original mass, or the original count rate.

Examples

1.

A sample of the radioisotope thorium-234 contains 8.0×10^{12} nuclei. The half-life of thorium-234 is 24 days. How many thorium-234 nuclei will remain in the sample after 120 days?	
Thinking	**Working**
Calculate how many half-lives 120 days corresponds to.	$n = \frac{120}{24}$ $= 5$ half-lives
Substitute $N_0 = 8.0 \times 10^{12}$ and $n = 5$ into the equation. Calculate the number of nuclei remaining.	$N = N_0 \frac{1}{2}^{n}$ $= 8.0 \times 10^{12} \times \frac{1}{2}^{5}$ $= 2.5 \times 10^{11}$ nuclei

2. A radioactive element X has a half-life of 2.00×10^3 years. If a 50.0 g sample gives a count rate of 1.00×10^3 counts per minute now, what would be:

(a) the count rate, and
(b) the mass of element left after 1.00×10^4 years?

$$\frac{1.00 \times 10^4}{2.00 \times 10^3} = 5 \text{ half - life periods}$$

$$\Rightarrow \quad n = 5$$

(a)
$$N = N_0 \frac{1}{2^n}$$
$$= (1.00 \times 10^3)(\frac{1}{2^5})$$
$$= 31.25 \text{ counts/min}$$

(b)
$$N = N_0 \frac{1}{2^n}$$
$$= (50.0)(\frac{1}{2^5})$$
$$= 1.562 \text{ g}$$

$\therefore$ The count rate is 31.2 counts/min and the mass left is 1.56 g.

Notes

1. The fact that only 1.56 g of element X is left in the example above, it does not mean that this is the mass of the sample. X would have transmuted into another element that would also be present, and hence would add its own mass to that of the remaining X.

   i.e. The 50.0 g of X would become 1.56 g of X and about 44.0 g of other elements, allowing for some mass being converted into energy during the transmutation process.

2. This relationship can be expressed mathematically as:

$$\boxed{N = N_0 e^{-\lambda t}}$$

where N = number of undecayed atoms at time t.
N_o = number of atoms present i n the original Sample
λ = radioactive constant for the particular element (the fraction of nuclei that decay per unit time)

This expression shows two things about radioactive decay.

- The number of radioactive atoms decreases with time.
- The level of radiation will approach zero, but never reach it.

Examples

1. A sample of the radioisotope thorium-234 contains 8.0×10^{12} nuclei. The half-life of ^{234}Th is 24 days. How many thorium-234 atoms will remain in the sample after:
   a 24 days?
   b 48 days?
   c 96 days?

   Solution

   a Initially, there were 8.0×10^{12} ^{234}Th nuclei. 24 days is one half-life, so half of these will decay, leaving 4.0×10^{12} ^{234}Th nuclei.
   b 48 days is two half-lives. This means that there will be $\frac{1}{2} \times \frac{1}{2} \times 8.0 \times 10^{12} = 2.0 \times 10^{12}$ nuclei.
   c 96 days corresponds to four half-lives. In this time the number of atoms of the original radioisotope will have halved four times. This means that $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$ or one-sixteenth of the original ^{234}Th nuclei remain, i.e. 5.0×10^{11} atoms.

2. In 2 hours, the activity of a sample of a radioactive element falls from 240 to 30 Bq. What is the half-life of this element?

   Solution

   During each half-life, the activity of the radioisotope will fall by half. The activity of this element has decreased from 240 → 120 → 60 → 30 counts per second, so it has decayed through three half-lives in this 2-hour (120-minute) period. Thus the half-life must be 120/3 = 40 minutes.

APPLICATIONS OF RADIOISOTOPES

Radioactive isotopes are widely used in the areas of industry, medicine and research. The usefulness of isotopes is related to their properties and to the effects of their radiation.

- ***Radiation can alter the structure of materials.***

  A single alpha or beta particle can cause the ionisation of several atoms as it passes through a substance. This ionising effect makes possible the operation of such devices as smoke detectors and the destruction of cancerous cells using radiation therapy.

- ***Radiation reveals information about the materials it interacts with.***

  The penetrating power of the different forms of radiation varies with the type of material. The amount of radiation passing through or reflected by a material can indicate its density or thickness. This can be used to discriminate between different materials or detect flaws within materials.

- ***Radioactive materials can be easily traced as they move through a system.***

  Even very small quantities of radioactive material can be detected fairly readily. This allows for their use in tracer techniques. Hence radioisotopes can be used to trace the path of chemicals within the body monitor biological processes or detect the occurrence of a leak from a gas pipeline.

Medical Uses Of Radioisotopes

Some of the most valuable applications of the use of radioisotopes occur in the field of medicine. Radioisotopes are powerful tools for analysis, diagnosis and treatment. Two important applications are radiation therapy and tracer techniques.

- ***Gamma radiation therapy*** involves the use of a high activity cobalt-60 source for the treatment of inoperable tumours.

  The source provides a thin beam of radiation, which is directed at the tumour. The source is continually rotated around the patient so that surrounding tissue will receive a relatively small exposure. The tumour, however, being the centre of this rotation, will receive a high dose of radiation and its cells will die.

- ***Tracer techniques.*** A radioactive isotope may be given to a patient by injection or swallowing and then its progress monitored as it moves through the body.

  Different substances are taken up by different organs of the body. Hence specific tracers are chosen to look at particular organs. (See the table below.)

  Radioisotopes chosen as tracers usually have a fairly short half-life. Since only very small amounts of tracer are introduced into the body the short half-life ensures a high activity and easy detection. It also means that the isotope is not active in the body for too long.

TABLE 3.4.1 Some common radioisotopes, their half-lives and applications.

Isotope	Emission	Half-life	Application
Natural			
polonium-214	α	0.00016 s	nothing at this time
strontium-90	β	28.8 years	cancer therapy
radium-226	α	1630 years	once used in luminous paints
carbon-14	β	5730 years	carbon dating of fossils
uranium-235	α	700000 years	nuclear fuel, rock dating
uranium-238	α	4.5 billion years	nuclear fuel, rock dating
thorium-232	α	14 billion years	fossil dating, nuclear fuel
Artificial			
technetium-99m	β	6 h	medical tracer
sodium-24	β	15 h	medical tracer
iodine-131	β	8 days	medical tracer
phosphorus-32	β	14.3 days	medical tracer
cobalt-60	β	5.3 years	radiation therapy
americium-241	α	460 years	smoke detectors
plutonium-239	α	24000 years	nuclear fuel, rock dating

The radioactive isotope technetium-99m is the most commonly used tracer. Its advantages include a short half-life (6.0 h), low production cost and chemical versatility. It can be introduced in various forms so as to be a tracer for checking such things as blood flow, thyroid function, bone disease, liver and heart function and detection of tumours.

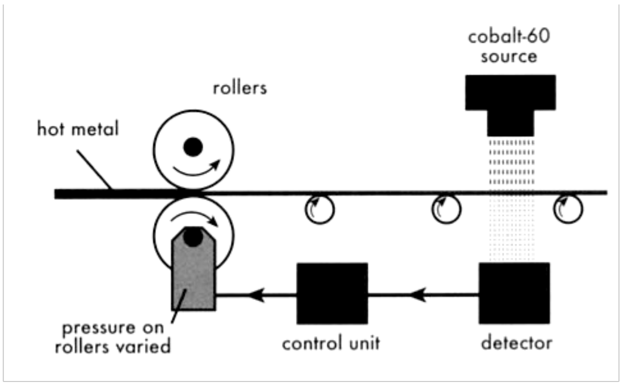
Other Uses Of Radioactive Isotopes

The use of radioisotopes in the areas of industry, agriculture and research are both interesting and varied. A brief description of some of these uses is listed below.

- ***Industrial radiography***. Isotopes such as iridium-192 can be used to check flaws in steel welding. This is much like an X-ray. The γ radiation will discriminate between materials of different density; the steel being denser than the weld. This technique of non-destructive quality control can also be used to inspect castings, ceramics and critical machinery such as aircraft engines.

- ***Gauging***. The thickness of sheet materials, such as sheet plastic or sheet metal, can be controlled without them being physically measured. The γ radiation from a source such as cobalt-60 has its intensity reduced as the thickness of the material it passes through increases.

  The reading by a radiation detector on the other side of the material can be used to monitor the correct thickness required during production.



In the packaging industry, the same technique can be used to ensure containers are correctly filled.

- ***Forensic analysis.*** Neutron activation analysis (NAA) is a quick and sensitive method for determining the proportion of elements that exist in any given sample. This is achieved by placing the sample in a reactor and then analysing the radiation of the artificial isotopes produced. The original source of the sample, perhaps from the scene of a crime, can then be established.

- ***Tracing applications***. These are many and varied. The flow rates of liquids and gases in pipelines and the flow rate of streams can be measured accurately using sodium-24 or cobalt-60 as a tracer. Importantly, leaks and blockages that occur in pipelines can also be investigated.

  e.g. Gold-198 has been used in a study to track Sydney's sewage after its disposal out to sea. The flow of hot materials inside a blast furnace can also be analysed for efficiency using this same isotope.
  In agriculture, tracers are important in the study of such things as biological processes, soil erosion, use of fertilisers and insect biology.

- ***Smoke detectors***. This important safety device is able to operate due to the ionising effect of americium-241, which emits a particles. An electronic alarm is prevented from sounding while a current is able to flow due to the presence of ionised air particles (see diagram). Smoke particles interfere with this current flow and hence trigger the alarm.

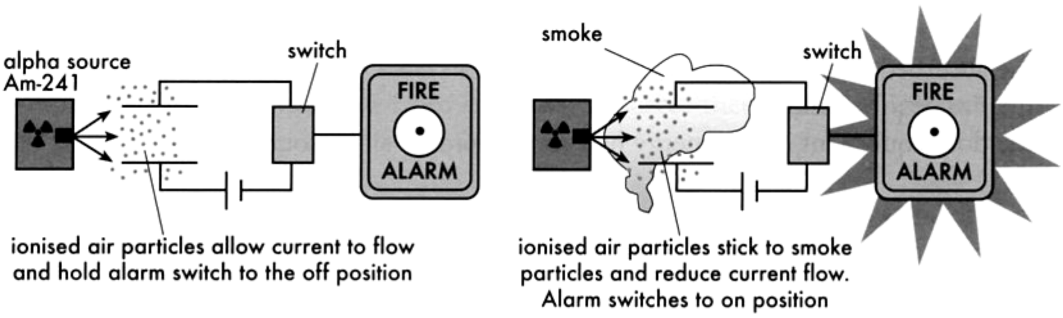


Figure 2.14 Operation of a smoke detector.

- ***Food irradiation.*** Cobalt-60 can be used to irradiate packaged food and help increase its shelf life. The γ radiation passes through the food, destroying bacteria and micro-organisms that causes food spoilage or food poisoning. Many countries do this, but not in Australia.

Carbon-14 Dating

C-14 has a half-life of 5730 years and can be used to determine the age of formerly living plants and animals. It is useful up to around 50,000 year old materials.

(*See the half-life diagram on the next page*.)

It is used in dating things such as bone, cloth, wood and plant fibres that were created in the relatively recent past by human activities.

Physics in action — Radiocarbon dating

Carbon dating is a technique used by archeologists to determine the age of fossils and ancient objects that were made from plant matter. In this method, the proportion of two isotopes of carbon, carbon-12 and carbon-14, in the specimen are measured and compared.

Carbon-12 is a stable isotope whereas carbon-14 is radioactive. Carbon-14 only exists in trace amounts in nature. In fact, carbon-12 atoms are about 1 000 000 000 000 (10^{12}) times more prevalent than carbon-14 atoms.

Carbon-14 has a half-life of 5730 years and decays by beta emission to nitrogen-14. Its decay equation is:

$$^{14}_{6}C \rightarrow ^{14}_{7}N + ^{0}_{-1}\beta$$

Both carbon-12 and carbon-14 can combine with other atoms in the environment, for example with oxygen to form carbon dioxide. While plants and animals are alive, they take in carbon-based molecules and so all living things will contain the same percentage of carbon-14. In the environment, the production of carbon-14 is matched by its decay and so the proportion of carbon-14 atoms to carbon-12 remains constant.

After a living thing has died, the amount of carbon-14 will decrease as these atoms decay to form nitrogen-14, and are not replaced. The number of atoms of carbon-12 does not change as this is a stable atom. So, over time, the proportion of carbon-14 to carbon-12 atoms falls. By comparing the proportion of carbon-14 to carbon-12 in a dead sample with that found in living things, and knowing the half-life of carbon-14 (5730 years), the approximate age of the specimen can be determined.

Consider this example. The count rate from a 1 g sample of carbon that has been extracted from an ancient wooden spear is 10 MBq. A 1 g sample of carbon from a living piece of wood gives a count rate of 40 MBq. For its count rate to have reduced from 40 to 10 MBq (40 → 20 → 10), the spear must be two half-lives of carbon-14 old, i.e. about 11 500 years old.

In 1988, scientists used carbon-dating techniques to show that the Shroud of Turin was probably a medieval forgery. Carbon-dating tests on samples of the cloth the size of a stamp established that there was a high probability that it was made between 1260 and 1390 AD, not around the time of Christ.

Figure 5.20

Carbon-dating techniques were used to show that the Shroud of Turin was most probably made around the 14th century.

More recent tests have cast doubt on the age of the shroud. Its age has again become a controversy.

Radiocarbon dating is an important aid to anthropologists who are interested in finding out about the migration patterns of early peoples—including the Australian Aboriginal people. This technique is very powerful since it can be applied to the remains of ancient campfires.

Rock Dating

Radioisotopes with very long half-lives are useful to determine the age of rocks, which have ages up to 4.5 billion years. (***See half-life diagram below.)***

Physics in action — Decay series

Generally, when a radionuclide decays, its daughter nucleus is not completely stable, and is itself radioactive. This daughter will then decay to a granddaughter nucleus which may also be radioactive, and so on. Eventually a stable isotope is reached and the sequence ends. This is known as a decay series.

The Earth is 4.5 billion years old (4.5 giga-years)—enough to have only four naturally occurring decay series remain active. These are:

- the uranium series in which uranium-238 eventually becomes lead-206
- the actinium series in which uranium-235 eventually becomes lead-207
- the thorium series in which thorium-232 eventually becomes lead-208
- the neptunium series in which neptunium-237 eventually becomes bismuth-209. (Since neptunium-237 has a relatively short half-life, it is no longer present in the crust of the Earth, but the rest of its decay series is still continuing.)

Geologists analyse the proportions of the radioactive elements in a sample of rock to gain a reasonable estimate of the rock's age. This technique is known as rock dating.

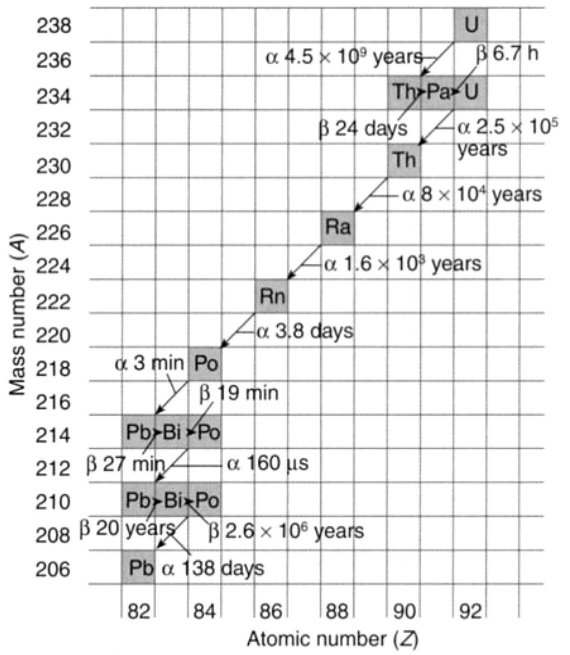


Figure 5.21
The uranium decay series. The half-life and emissions are indicated on each of the decays as radioactive uranium-238 is transformed into stable lead-206.

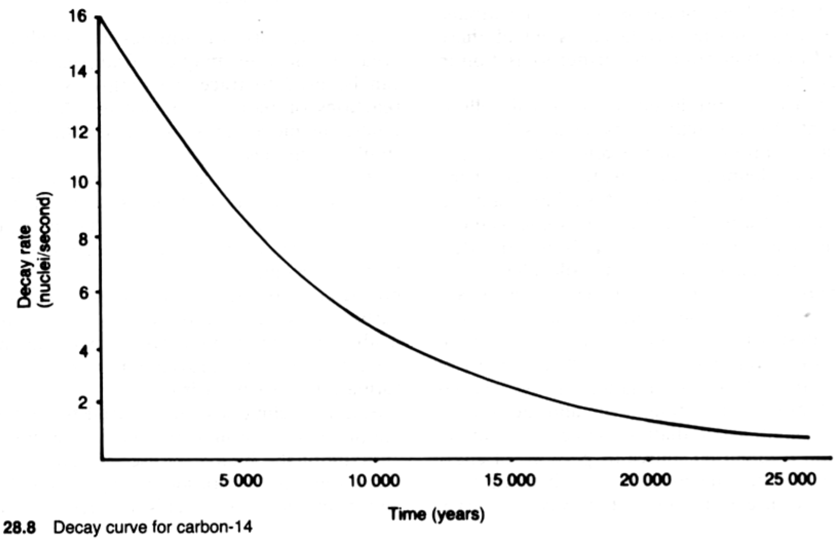


Figure 28.8 Decay curve for carbon-14

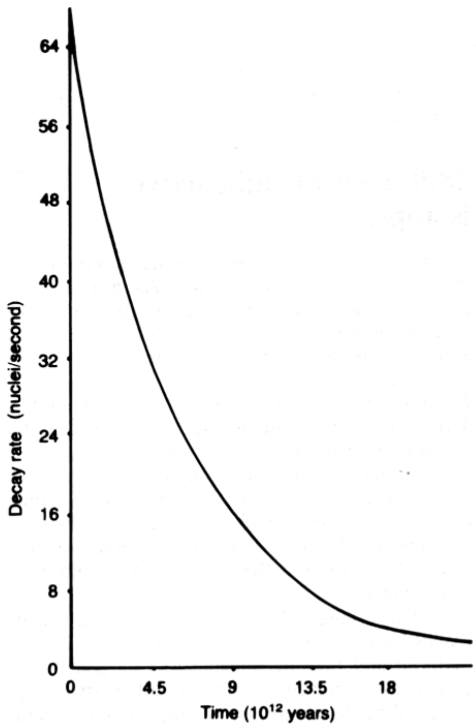


Figure 28.9 Decay curve for uranium-238

BIOLOGICAL EFFECTS OF RADIATION

Radiation damage to the living body cells depends on two factors.

- The ***energy*** the cells absorb from the radiation.
- The ***time*** over which the cells absorb the radiation energy.

When ionising radiation interacts with living cells it transfers its energy to the individual atoms that make up the cell. This may cause damage to the cell by ionisation and the formation of extremely chemically reactive atoms or groups of atoms known as ***free radicals.***

.

Where atoms or molecules are ionised, this can cause a change in the cell processes and cell chemistry.

e.g. If DNA molecules are ionised, the effect may be the prevention of cell division , the premature death of the cell or a permanent genetic modification.

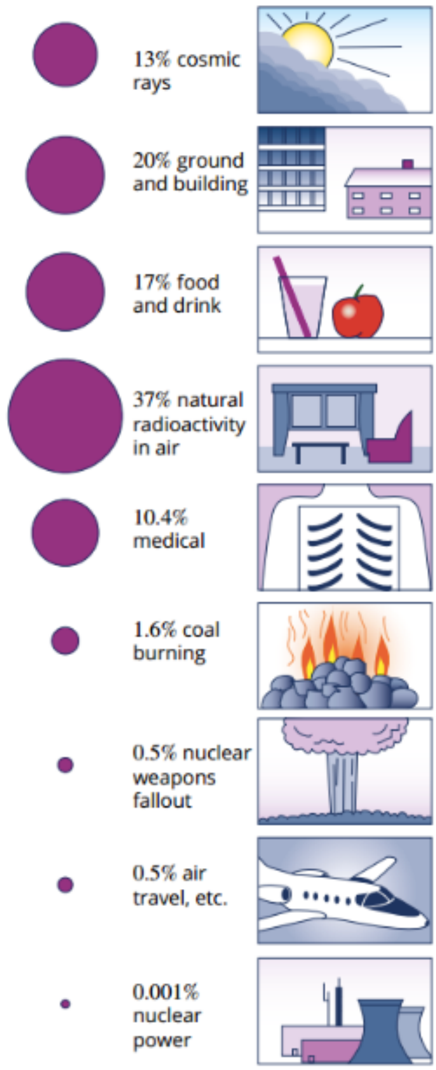


FIGURE 3.5.4 Everyone is exposed to radiation from the environment.

Radiation Dose

The energy that our bodies receive from ionising radiation is referred to as the ***absorbed dose***. Specifically, the absorbed dose is ***the energy absorbed per kilogram of tissue***.

$$\boxed{\text{Absorbed dose} = \frac{\text{energy absorbed}}{\text{mass of tissue}} = \frac{E}{m}}$$

The SI unit for absorbed dose is the gray (Gy).

$$1.00 \text{ Gy} = 1.00 \text{ Jkg}^{-1}$$

The Quality Factor

Absorbed dose is useful when comparing the effects of different amounts of the same radiation. However, it is not useful when the effects of different types of radiation are considered.

The effects of different types of radiation on living cells vary. Alpha particles are far more ionising than gamma rays because of their large size. Hence equal doses of these two types of radiation can have quite different biological effects.

The ***quality factor*** for a particular radiation indicates the cell damage it causes in comparison to an equal dose of γ rays. The ***dose equivalent*** (in Sieverts) is a measurement that incorporates the quality factor.

This means that 1.00 Sievert of any type of radiation will have the same biological effect.

Dose equivalent is given by the following.

$$\boxed{\text{Dose equivalent} = \text{Absorbed dose x Quality factor}}$$

1.00 sievert (Sv) = 1.00 Jkg^{-1}

Quality Factors

Radiation Type	Approximate Quality Factor
gamma rays	1
beta particles	1
slow neutrons	3
fast neutrons	10
alpha particles	20

Examples

1. A person of mass 70.0 kg accidentally absorbs 45.0 J of energy from a beta radiation source. Determine:

   (a) the absorbed dose.
   (b) the dose equivalent.
   (c) the dose equivalent if this energy had come from an alpha source.

   (a)

$$\begin{aligned}\text{Absorbed dose} &= \frac{\text{energy absorbed}}{\text{mass of tissue}} \\ &= \frac{45.0}{70.0} \\ &= \underline{0.643\ \text{Gy}}\end{aligned}$$

   (b)

$$\begin{aligned}\text{Dose equivalent} &= \text{Absorbed dose x QF} \\ &= (0.643)(1) \\ &= \underline{0.643\ \text{Sv}}\end{aligned}$$

   (c)

$$\begin{aligned}\text{Dose equivalent} &= \text{Absorbed dose x QF} \\ &= (0.643)(20) \\ &= \underline{12.9\ \text{Sv}}\end{aligned}$$

2.

DOSE EQUIVALENT

Calculate the dose equivalent (in μSv) from various types of radiation if the absorbed dose is 0.50 μGy.	
Thinking	**Working**
The quality factor for each type of radiation can be found in Table 3.5.1 (page 77). $1\,\mu Gy = 1 \times 10^{-6}\,Gy$	QF (alpha particles) = 20 QF (beta particles) = 1 QF (gamma rays) = 1

a Calculate the dose equivalent (in μSv) if the source is emitting alpha particles.	
The dose equivalent DE = AD × QF.	$DE(\alpha) = 0.50 \times 10^{-6} \times 20$ $= 1 \times 10^{-5}\,Sv$ $= 10\,\mu Sv$

b Calculate the dose equivalent (in μSv) if the source is emitting beta particles.	
Thinking	**Working**
The dose equivalent DE = AD × QF.	$DE(\beta) = 0.50 \times 10^{-6} \times 1$ $= 0.50 \times 10^{-6}$ $= 0.50\,\mu Sv$

c Calculate the dose equivalent (in μSv) if the source is emitting gamma rays.	
Thinking	**Working**
The dose equivalent DE = AD × QF.	$DE(\gamma) = 0.50 \times 10^{-6} \times 1$ $= 0.50 \times 10^{-6}$ $= 0.50\,\mu Sv$

3.

TREATING TUMOURS

A 10 g cancer tumour absorbs 2.5×10^{-3} J of energy from an applied radiation source. Calculate the dose equivalent if the source is an alpha emitter using information from Table 3.5.1.	
Thinking	**Working**
Convert mass from grams to kg.	$m = \frac{10}{1000}$ $= 0.010\,kg$
Calculate the absorbed dose (AD) using the energy and the mass.	$AD = \frac{E}{m}$ $= \frac{2.5 \times 10^{-3}}{0.010}$ $= 0.25\,Gy$
Calculate the dose equivalent (DE) using the quality factor for alpha particles of 20.	$DE = AD \times QF$ $= 0.25 \times 20$ $= 5\,Sv$

Effects Of Ionising Radiation On People

The effect of radiation on our bodies is due to the damage or changes that occur at the cellular level. The magnitude of this effect is proportional to the radiation dose equivalent received. The symptoms from a large radiation dose to the body become evident within a few days or weeks. These include nausea and vomiting and a general unwellness (radiation sickness).

The likely effects of large radiation doses are:

- 1,000 mSv - nausea, vomiting and diarrhoea (radiation sickness) will occur within days. Fatal cancers may form later in life.
- 10,000 mSv - death likely to result within 2 weeks due to damage to white blood cells and the gastrointestine.
- 100,000 mSv - death within hours or days due to severe damage to the central nervous system.

Some parts of the body are more sensitive to ionising radiation than others. The most sensitive are bone marrow, the reproductive organs, the eye and the digestive and circulatory systems.

e.g. Doses of 2000-3000 mSv to the testes or ovaries will cause permanent sterility while similar doses to the eye will result in cataracts.

Lower doses of radiation may produce effects that only become apparent after several months or even years. These include a reduced life expectancy, the development of leukaemia or tumours and cataracts on the eye lens. Genetic damage, if it occurs, may not be apparent for several generations.

Dose Equivalent (Sv)	Body Damage
0.25	Reduction in lymphocyte (white blood cell) count.
1	Possible nausea, vomiting.
4	Nausea, diarrhoea, drop in blood cell count. About 50% of exposed group die within weeks from failure of blood-forming organs.
5	Loss of hair, 50% die.
6	Damage to stomach and intestine walls with loss of fluids. Immediate radiation sickness, bloody diarrhoea, extreme thirst, purpura, death within 3 weeks.
10	Severe damage to central nervous system. Death within days.

Lethal Dose

LD is the symbol for lethal dose, meaning a dose that results in death.

LD-50 represents a mean lethal dose that will kill 50% of the exposed people.

This has been extended to LD-50/25, which represents the dose that will kill 50% of exposed people in 25 days.

Background radiation

It is important to appreciate that 1.00 Sv is a massive dose of radiation. It would not be fatal, but it would certainly lead to a severe case of radiation sickness.

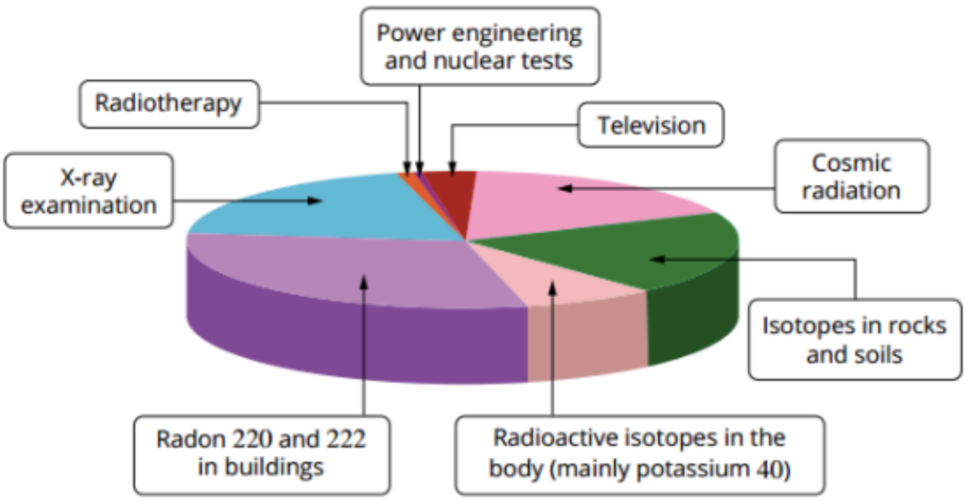
In Australia, the average annual background radiation dose is about 2.0 mSv, or 2000μSv2000μSv.

Use the following table to estimate your dose over the past year.

TABLE 3.5.2 Annual radiation doses in Australia for different radiation sources.

Radiation source	Average annual dose (μSv)	Local variations
cosmic radiation	300	plus 200μSv for each round-the-world flight plus 20μSv for each 10° of latitude plus 150μSv if you live 1000 m above sea level
rocks, air and water	1350	plus 1350μSv if you live underground plus 1350μSv if your house is made of granite minus 140μSv if you live in a weatherboard house
radioactive foods and drinks	350	plus 1000μSv if you have eaten food affected by the Fukushima fallout
manufactured radiation	60	plus 60μSv if you live near a coal-burning power station plus 30μSv from nuclear testing in the Pacific

Share of the different sources of radiation
The average equivalent of the dose absorbed by a person in a year is equal to a few millisieverts.



Natural sources	*DE* (mSv)	Artificial sources	*DE* (mSv)
cosmic radiation	0.3–0.5	X-ray examination	0.5
isotopes in rocks and soils	0.4–0.5	radiotherapy	0.01
radioactive isotopes in the body (mainly potassium 40)	0.2	power engineering and nuclear tests	0.01
radon 220 and 222 in buildings	0.6–0.8	television	0.08
total	1.5–2.0	total	0.6

FIGURE 3.5.6 Pie chart showing common sources of background radiation.

Effect of Radiation Exposure on Humans

High levels of radiation exposure can have an adverse effect on human tissue.

Radiation exposure can be ***chronic*** (occurs over a long period of time), as expected by a worker in a nuclear power plant.

It can be ***acute*** (occurs all at once), such as in a nuclear reactor accident or explosion.

Immediate effects of radiation exposure include a drop in the white blood cell count, nausea, fatigue, hair-loss and skin reddening. Radiation has the potential to cause DNA damage and cause cells to become cancerous.

In addition, different radioactive elements can target different parts of the body.

e.g. Iodine affects the thyroid and strontium-90 affects bone and bone marrow.

TABLE 3.5.3 Effect of radiation dosage on the human body.

Radiation doses	Effects
1.5 mSv	Typical background exposure in Australia
9 mSv/year	Exposure by airline crew on the New York–Tokyo route
100 mSv/year	Highest annual safe level; above this the probability of cancer is assumed to increase with the dose 4 months on the International Space Station, 350 km above the Earth
350 mSV/year	Criterion for relocating people after the Chernobyl accident
700 mSv/year	Suggested threshold for maintaining evacuation after a nuclear incident
1000 mSv short term, whole body	Threshold for causing radiation sickness and nausea, but not death
5000 mSv whole body	Fatal for 50% of those exposed
10 000 mSv whole body	Acute radiation poisoning, death within a few weeks

Radiation safety

The fact that we cannot see radiation makes it essential that appropriate protective safety measures are taken. Regulations exist that cover the safe handling, storage and waste disposal of radioactive materials.

The preventative measures necessary depend on the type of radiation and the strength of the source.

To minimise the dose from radioactive sources there are three important factors to consider. These are:

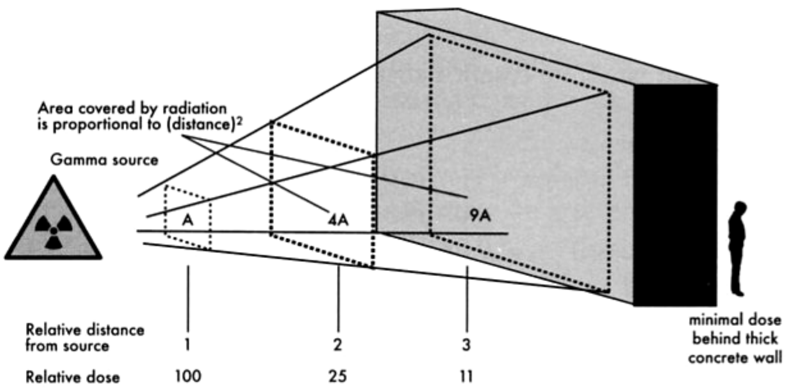
- ***Distance*** - the further you are from a source the safer you are. Not only is some of the radiation absorbed by the air or materials between you and the source but more importantly the intensity will drop off rapidly with distance. (See the diagram below.)

- ***Shielding*** - is particularly important if distance cannot be minimised. It can be chosen to absorb particular radiation.

  e.g. Perspex sheets will stop beta radiation; lead and concrete will absorb gamma rays, while boron and cadmium are good absorbers of neutrons.

- ***Time*** - the longer you are exposed to a source the greater the absorbed dose. Good planning can minimise this.

  e.g. Radioisotopes used in medicine are chosen so that they have a short biological half-life. This means that they are present in the body for a shorter time.



The effect of distance on absorbed dose. The intensity of the radiation is proportional to ($1/d^2$).

NUCLEAR BINDING ENERGY

When a nucleus is bombarded with a suitable particle, it may cause a nuclear reaction if its E_k is sufficiently large to overcome the energy that is binding the nuclear particles together. If a reaction occurs, the nucleus is rearranged into a more stable configuration with a release of some energy.

> ***Binding Energy***: the energy that must be supplied by a bombarding particle to separate the nucleus into its component parts.

The standard unit of atomic mass is the ***atomic mass unit (u).***

$$\boxed{\begin{aligned} 1.00\text{ u} &= {}^{1}/_{12} \text{ x mass of one C-12 atom} \\ &= 1.66 \text{ x } 10^{-27} \text{ kg} \end{aligned}}$$

The energy equivalent of this is given by Einstein's Equation.

$$\begin{aligned} E &= mc^2 \\ &= (1.66 \text{ x } 10^{-27})\,(3.00 \text{ x } 10^{8})^2 \\ &= 1.494 \text{ x } 10^{-10} \text{ J} \\ &= 9.314 \text{ x } 10^{8} \text{ eV} \\ &= 931.4 \text{ MeV} \end{aligned}$$

$1.00 \text{ eV} = 1.60 \text{ x } 10^{-19}$ J

$\therefore$ Divide by $1.60 \text{ x } 10^{-19}$

$$\boxed{\therefore \quad 1.00 \text{ u} = 931.4 \text{ MeV}}$$

Note The binding energy is usually measured in MeV.

When individual particles come together to form a nucleus, energy is released. This release of energy is always accompanied by a loss of mass. This loss of mass is often referred to as the ***mass defect***.

This means that the measured mass of a nucleus is always less than the sum of the individual masses of its nucleons.

Examples

1. Determine the binding energy per nucleon for the Helium atom.

   A Heliurn atom is made up of 2 protons., 2 neutrons and 2 electrons. Hence its mass should be:

$$\begin{aligned} 2 \text{ x } m_p &= 2(1.00728) = 2.01456 \text{ u} \\ 2 \text{ x } m_n &= 2(1.00867) = 2.01734 \text{ u} \\ 2 \text{ x } m_e &= 2(0.00055) = \underline{0.001113 \text{ u}} \\ & \qquad\qquad\qquad\quad 4.03300 \text{ u} \end{aligned}$$

   The mass of a Helium atom is 4.00389 u.

$$\begin{aligned} \therefore \quad \text{Mass defect} &= 4.03300 \text{ u} - 4.00389 \text{ u} \\ &= 0.02911 \text{ u} \\ &= (0.02911)(931.4) \\ &= 27.11 \text{ MeV} \end{aligned}$$

Helium has 4 nucleons.

$\therefore$ Binding energy per nucleon $= \dfrac{27.11}{4}$

$= 6.778$ MeV

$\therefore$ <u>The binding energy per nucleon = 6.80 MeV</u>

Note: The binding energy for the nucleus is 27.11 MeV. This is the energy a bombarding particle needs to split the nucleus into its constituent parts.

2. ${}^{232}_{92}U$ decays to ${}^{228}_{90}Th$ by emitting an alpha particle.

(a) What energy is released when this disintegration takes place?
(b) In what form is this energy released?

Masses ${}^{232}_{92}U$ = 232.0372 u
${}^{228}_{90}Th$ = 228.0207 u
${}^{4}_{2}He$ = 4.0039 u

$${}^{232}_{92}U \rightarrow {}^{228}_{90}U + {}^{4}_{2}He + \text{energy}$$

$\Rightarrow$ 232.0372 u $\rightarrow$ 228.0287 u + 4.0039 u
$\Rightarrow$ 232.0372 u > 232.0326 u

Mass difference = mass of reactants - mass of products
= 232.0372 u - 232.0326 u
= 0.0046 u
= (0.0046)(931.4)
= 4.284 MeV

$\therefore$ <u>Hence the energy released is 4.3 MeV, mainly as the E_k of the alpha particle.</u>

Note It is possible to do "mass defect" problems with masses in kilograms. This requires converting Joules to electron volts.

3.

BINDING ENERGY

Calculate the average binding energy per nucleon for the carbon-12 nucleus in MeV and joules. Use the following data in your calculations: mass of a carbon-12 nucleus = 11.993 417 u, mass of a proton = 1.007 276 u and mass of a neutron = 1.008 664 u.	
Thinking	**Working**
Determine the total mass of the nucleons in a carbon-12 nucleus.	total mass = mass of 6 neutrons + mass of 6 protons $= (6 \times 1.008664) + (6 \times 1.007276)$ $= 12.095640$ u
Determine the mass defect.	mass of nucleons − actual mass of nucleus $= 12.095640 - 11.993417$ $= 0.102223$ u
Determine the binding energy in MeV.	binding energy = mass defect × 931 MeV $= 0.102223 \times 931$ $= 95.17$ MeV
Determine the binding energy per nucleon in MeV.	$= \frac{95.17}{12}$ $= 7.93$ MeV per nucleon
Determine the binding energy per nucleon in J.	$= 7.93 \times 10^{6} \times 1.60 \times 10^{-19}$ $= 1.27 \times 10^{-12}$ J

BINDING ENERGY AND MASS NUMBER

The binding energy of heavier nuclei is larger than that for light nuclei due to the greater number of nucleons that are separated.

The binding energy per nucleon increases as the mass number approaches 60. Atoms with a mass near this number are most stable because they have the highest binding energy per nucleon.

After 60, the binding energy per nucleon decreases, indicating a decrease in nucleus stability as more nucleons are packed in.

Generally, nuclei heavier than 60 will lose energy by disintegrating (in a process called ***fission***), and those below 60 will lose energy by combining with other nuclei (in a process called ***fusion***). In both cases, the nuclei will move towards the most stable mass number of 60.

Note Some lighter nuclei such as He-4, C-12 and O-16 have a nuclear stability that does not fit the general case.

The following graph shows the general trend in binding energy per nucleon.

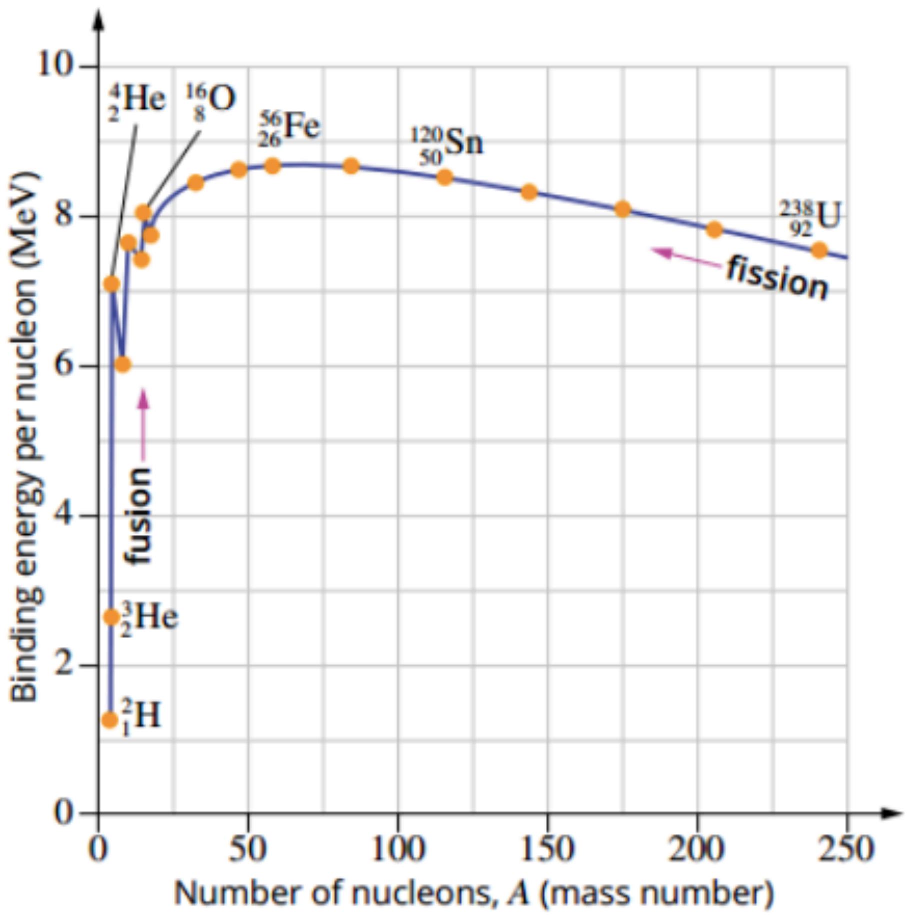


FIGURE 4.3.6 The graph of binding energy per nucleon.

NUCLEAR FISSION

Nuclear fission is the splitting of the nucleus of an atom into two smaller nuclei. In this process large amounts of energy are released as well as gamma radiation and some neutrons . Fission can be spontaneous or induced by the absorption of neutrons.

A typical fission reaction for uranium-235 is:

$$ {}^{1}_{0}n \; + \; {}^{235}_{92}U \; \rightarrow \; {}^{91}_{36}K \; + \; {}^{142}_{56}Ba \; + \; 3\,{}^{1}_{0}n \; + \; \text{energy} $$

Krypton-91 and barium-142 are known as the fission products or ***fission fragments***. Three neutrons are freed from the uranium nucleus when it splits.

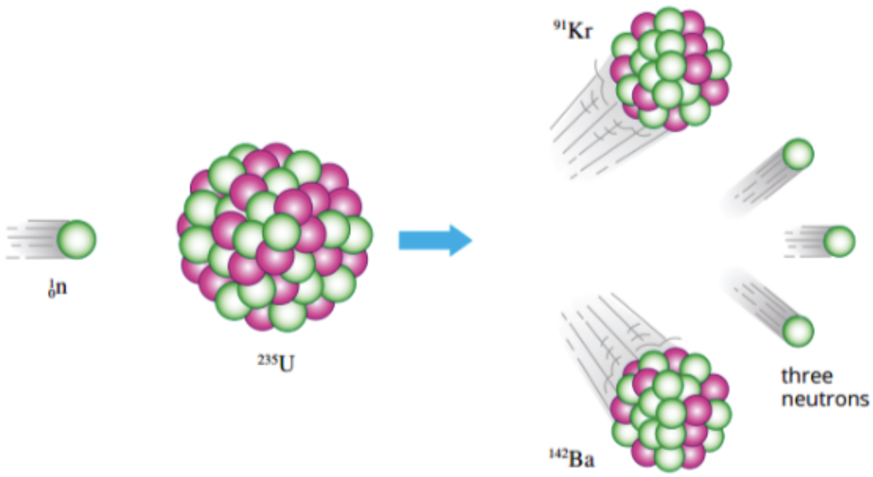


FIGURE 4.1.6 One possible outcome for the neutron-induced fission of uranium-235.

The uranium can break up into any of several different combinations of nuclides. Typically, however, the heavier particle has a mass number around 130 to 150 and the lighter fragment around 90 to 100.

Over 40 different pairs of fission fragments of uranium-235 have been found, and most of these are radioactive ***beta emitters***. It is these ***radioactive fission fragments*** that comprise the bulk of the ***high level waste*** produced by nuclear reactors.

Typically around 200 MeV is released during a fission reaction.

Example

FISSION

Plutonium-239 is a fissile material. When a plutonium-239 nucleus is struck by and absorbs a neutron, it can split in many different ways. Consider the example of a nucleus that splits into barium-145 and strontium-93 and releases some neutrons.

The nuclear equation for this is:

$${}^{1}_{0}n + {}^{239}_{94}Pu \rightarrow {}^{145}_{56}Ba + {}^{93}_{38}Sr + a{}^{1}_{0}n + \text{energy}$$

a How many neutrons are released during this fission process, i.e. what is the value of a?

Thinking	Working
Analyse the mass numbers (A).	$1 + 239 = 145 + 93 + (a \times 1)$ $a = (1 + 239) - (145 + 93)$ $= 2$ 2 neutrons are released during this fission.

b During this single fission reaction, there is a loss of mass (a mass defect) of 3.07×10^{-28} kg. Calculate the amount of energy that is released during the fission of a single plutonium-239 nucleus. Answer in both MeV and joules.

Thinking	Working
The energy released during the fission of this plutonium nucleus can be found by using $E = mc^2$.	$\Delta E = \Delta mc^2$ $= (3.07 \times 10^{-28}) \times (3.00 \times 10^8)^2$ $= 2.76 \times 10^{-11}$ J
To convert J into eV, divide by 1.6×10^{-19}. Remember that 1 MeV = 10^6 eV.	$E = \dfrac{2.76 \times 10^{-11}}{1.6 \times 10^{-19}}$ $= 1.73 \times 10^8$ eV $= 173$ MeV

c The combined mass of the plutonium nucleus and bombarding neutron is 3.99×10^{-25} kg. What percentage of this initial mass is converted into the energy produced during the fission process?

Thinking	Working
Use the relationship percentage of initial mass converted into energy $= \dfrac{\text{mass defect}}{\text{initial mass}} \times \dfrac{100}{1}$	percentage of initial mass converted into energy $= \dfrac{\text{mass defect}}{\text{initial mass}} \times \dfrac{100}{1}$ $= \dfrac{3.07 \times 10^{-28}}{3.99 \times 10^{-25}} \times \dfrac{100}{1}$ $= 0.0769\%$ This is a very small percentage loss in mass.

PHYSICS IN ACTION

Enrico Fermi, Lise Meitner and nuclear fission

Enrico Fermi, pictured in Figure 4.1.9, was born in Italy in 1901. He completed his doctorate and post-doctorate work in physics at the University of Pisa and in Germany. Fermi had emigrated to the USA by the time the nuclear age dawned in the 1930s. The neutron had just been discovered in 1932, which enabled scientists to fire neutral particles at atomic nuclei for the first time. Fermi was at the forefront of this research.

Fermi bombarded uranium-238 atoms with neutrons and found that uranium-238 nuclei absorbed the neutrons and formed a radioactive isotope of uranium. This isotope then decayed by emitting a beta-minus particle to become neptunium, which then emitted another beta-minus particle to become plutonium, two completely undiscovered elements. Fermi had successfully produced the world's first artificial and **transuranic** (i.e. after uranium) elements. The nuclear reactions for this process are:

$${}^{1}_{0}\text{n} + {}^{238}_{92}\text{U} \rightarrow {}^{239}_{92}\text{U}$$

$${}^{239}_{92}\text{U} \rightarrow {}^{239}_{93}\text{Np} + {}^{0}_{-1}\beta$$

$${}^{239}_{93}\text{Np} \rightarrow {}^{239}_{94}\text{Pu} + {}^{0}_{-1}\beta$$

In 1938, following on from Fermi's work, two German scientists, Otto Hahn and Fritz Strassmann, were also bombarding uranium ($Z = 92$) in an attempt to produce some transuranic elements ($Z > 92$). They found that, rather than producing larger elements, they were getting isotopes of barium ($Z = 56$). Hahn wrote to his colleague Lise Meitner, pictured in Figure 4.1.10, about this unexpected result. She then discussed this with her nephew Otto Frisch, a nuclear physicist, and realised that the bombarding neutrons were causing the uranium nuclei to split.

If barium ($Z = 56$) was one of the products, then krypton ($Z = 36$) must be another. This was found to be the case. It was Frisch who coined the term 'fission' and Meitner who proposed that energy would be released during this process.

After the start of World War II, Enrico Fermi was commissioned by President Roosevelt to design and build a device that would sustain the fission process in the form of a chain reaction. In 1942, Fermi succeeded in this task. A squash court at the University of Chicago was used as the site for the world's first nuclear reactor. It produced less than 1 W of power—not even enough to power a small light globe! This sounds like a bit of a failure, but in fact, achieving fission for the first time was a very important breakthrough. The reactor was later modified to produce about 200 W. Fermi died of cancer in 1954. One year after his death, the element with atomic number 100 was artificially produced and named fermium, Fm, in his honour.

FIGURE 4.1.9 Enrico Fermi.

FIGURE 4.1.10 Lise Meitner.

Chain Reactions

When a uranium-235 nucleus undergoes a fission reaction , some neutrons, usually two or three, are always released . If at least one of these neutrons is able to hit another uranium-235 nucleus , a self- sustaining chain reaction results.

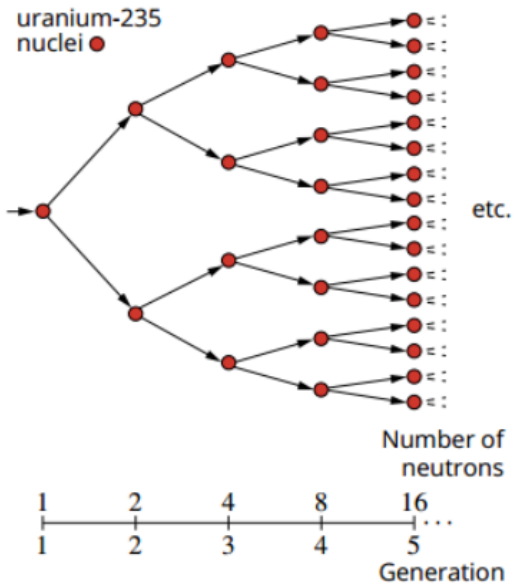


FIGURE 4.2.2 A chain reaction of nuclei.

In the diagram above, two neutrons are released during the fission reaction. The number of nuclei undergoing fission doubles each generation and within a small fraction of a second an enormous number of nuclei have undergone fission.

Only a minuscule amount of energy (of the order of 10^{-13} J) is released by each fission reaction, but in this uncontrolled chain reaction there are so many reactions occurring in such a short time that an explosion results.

In 1.00 kg of uranium-235, so many reactions occur that about 8 x 10^{13} J of energy is released in just over one-millionth of a second.

Fortunately, such a chain reaction does not easily occur due to a variety of factors.

- ***Neutrons may simply escape from the material without hitting any nuclei.*** Remember, nuclei are an extremely small part of an atom.

- ***Neutrons may be absorbed by uranium -238 nuclei, which rarely undergo fission***. Only 0.7% of naturally occurring uranium is the fissile uranium-235.

- ***Only the very slow neutrons, called thermal neutrons , are likely to cause uranium-235 to split***. Fast neutrons emitted from a fission reaction are more easily captured by uranium-238.

Critical Mass

Critical mass refers to the mass of fissile material required to sustain an ***uncontrolled chain reaction*** such as that which occurs in an atom bomb. If too little matter is present, most of the neutrons produced in a fission reaction escape from the material without interaction .

Critical mass depends on:

- ***the % of uranium-235 or plutonium-239 in a sample***. (Uranium-238 is not fissile.)

  The fuel used in nuclear fission weapons is enriched to a high degree of purity (over 90 % U-235)) so that a chain reaction can be sustained. Nuclear reactors require 3 % U-235.

- ***the geometry (shape) of the mass***.

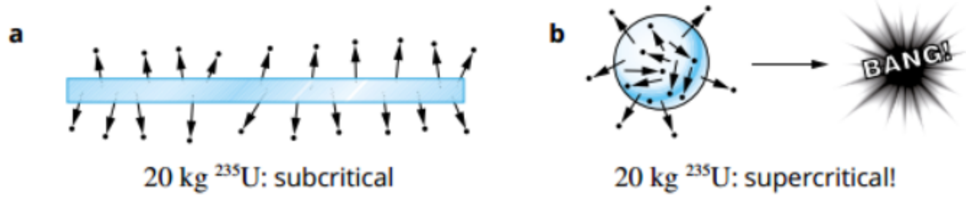


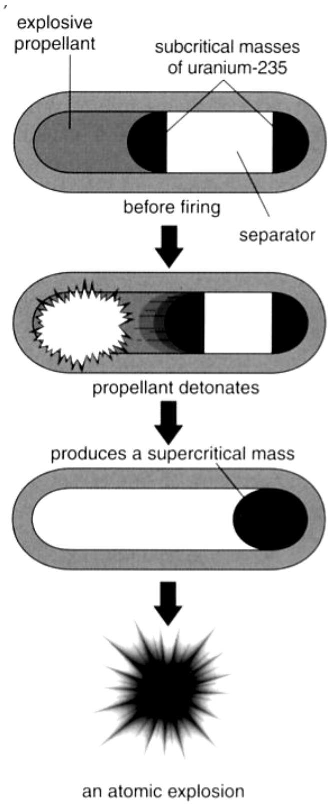
FIGURE 4.2.3 (a) A large proportion of neutrons escape the flat piece of uranium. (b) A sufficient proportion of neutrons remain to maintain the chain reaction—leading to an explosion.

Atomic Bomb

When an atom bomb is detonated, two separate masses of fissile material, each less than critical mass, are forced together by a chemical explosive.

The two combined masses are then greater than critical mass and an explosion results.

The nuclear bomb that was dropped on Hiroshima contained two hemispherical sub-critical pieces of 95% pure uranium-235. This bomb, known as Little Boy, was dropped by a B-29 bomber called the *Enola Gay* on 6 August 1945. When it reached an altitude of 580 m, an explosive charge fired one piece into the other, creating one supercritical mass of uranium- 235, which exploded within one-millionth of a second after this.



Thermal Nuclear Reactors

Nuclear reactors are able to produce large quantities of heat as a result of a controlled chain reaction. Enriched uranium-235 is used in thermal reactors while plutonium-239 is used in fast-breeder reactors. The heat produced in a reactor passes through a heat exchanger and eventually drives a turbine to produce electricity .

Reactor design can vary widely with the choice of fuel and moderator being of particular importance. In order that a controlled chain reaction can be sustained in a reactor an average of one neutron from each fission must cause another fission.

Hence enriched uranium is preferred since natural uranium only contains 0.7% of this fissile material. Also, since slow neutrons are more likely to cause the fission of uranium-235 a moderator is used to slow down the energetic neutrons produced during fission reactions .

The design features of thermal nuclear reactors include:

- ***Nuclear fuel*** Modem reactors use enriched uranium (≈ ***2 - 3 % uranium-235***) in the form of uranium dioxide.
- ***Moderator*** Although fast neutrons can cause fission in uranium-238 (the more abundant isotope), the probability of fission of uranium-235 by slow neutrons is much higher. Hence a moderator is used to deliberately ***slow down the neutrons*** to normal thermal energy. Hence the term thermal reactors. Graphite, water, heavy water and beryllium are commonly used as moderator materials.
- ***Control rods*** The rate of the fission reaction is controlled by lowering or raising the boron- steel control rods. ***Neutrons are absorbed*** by the boron and inhibit the chain reaction.
- ***Coolant*** This can be water, a gas (CO_2), or liquid metal (Na). The coolant ***removes heat from the reactor core*** and transfers it to the outside of the reactor for the generation of electricity.
- ***Radiation shield*** This consists of very thick reinforced concrete with steel, graphite and lead layers. Its ***purpose is to protect*** workers and the nearby environment.

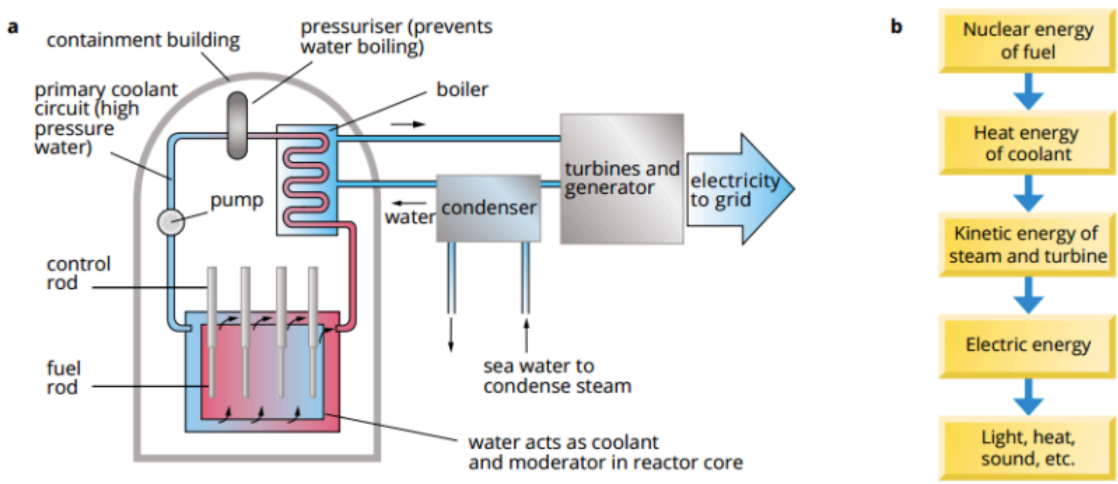


FIGURE 4.2.6 (a) A schematic diagram of a nuclear reactor; (b) energy transformations involved as a reactor is used to produce electricity.

Fast-Breeder Reactors

The Earth's known supply of the relatively scarce uranium-235 may run out within a few decades if its use is increased. In this event, many current nuclear reactors will become obsolete. A different type of reactor, known as a ***fast breeder reactor***, may therefore come into greater prominence in the future. Fast breeder reactors make use of the most abundant uranium isotope, uranium-238.

These reactors use ***mixed plutonium uranium fuel***, which is initially produced as a by-product of thermal reactors. Fast moving neutrons cause the fission of the plutonium-239 and hence the release of energy. These reactors also ***convert the normally wasted uranium-238 to plutonium-239***, hence the name breeder reactors.

The plutonium-239 is then extracted for use in the next generation of fuel rods. However, it can be extracted for use in atomic weapons and this makes their use highly controversial.

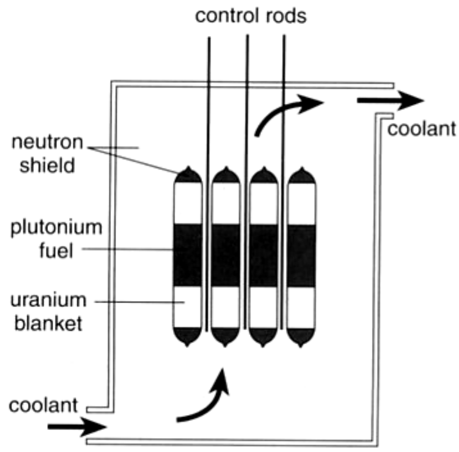


Figure 5.38
The core of a fast breeder reactor does not include a moderator. The plutonium in the fuel rods is surrounded by uranium. It is here that new plutonium is bred.

Fast breeder reactors are very expensive to build and they suffer from many technological difficulties. At present, there are only a handful of fast breeder reactors in operation in France, the United States, Britain, Russia and Japan.

PHYSICSFILE

Nuclear disasters: Fukushima

The biggest earthquake in Japan's history occurred on 11 March 2011 and triggered a massive tsunami along the east coast. Tragically, about 20000 people were killed by the tsunami.

Japan has 52 nuclear reactors. The Fukushima nuclear power plant, consisting of six reactors, was close to the coast and protected by sea walls designed to withstand a 5.7 metre tsunami. The 14 metre wave that arrived about an hour after the earthquake completely inundated the reactors, electrical switching systems and generators. The pumps that pushed water through the reactors stopped working, causing the reactors to overheat. Hydrogen explosions caused fires and damaged the containment vessels. This resulted in the release of radioisotopes iodine-131 and caesium-137 into the atmosphere. Water leaked from Reactor 1, leaving fuel rods exposed, thus triggering their meltdown. As pictured in Figure 4.2.7, one hydrogen explosion caused the roof of Fukushima's Reactor 1 to collapse, releasing radioactive materials into the atmosphere. The Japanese government created a 20 km exclusion zone around the power plant, forcing 80000 people from their homes.

Nuclear accidents are rated according to the International Nuclear and Radiological Event scale from 1 to 7, with 7 being the most serious. Both the Fukushima and Chernobyl disasters (which occurred in Ukraine in 1986) were rated at 7.

FIGURE 4.2.7 A hydrogen explosion caused the roof of Reactor 1 at Fukushima to collapse, releasing radioactive materials into the atmosphere.

NUCLEAR FUSION

An important type of nuclear reaction that occurs in nature is ***fusion***. Fusion occurs in the Sun and the stars. It is how heavier elements are formed from light nuclei such as hydrogen and helium.

Fusion involves the joining of two small nuclei to form a larger one. When this occurs, energy is released and the mass of the products is less than the mass of the reactants.

This release of energy is possible since atoms such as helium have a much greater binding energy per nucleon than smaller atoms such as hydrogen and deuterium. This indicates that the helium nucleus is particularly stable.

Fusion reactions are difficult to sustain. Strong electrostatic forces prevent nuclei from coming close enough together for the strong nuclear forces to be effective. Temperatures in the order of 100 million K (10^8 K) are necessary to maintain such reactions. This forms a plasma of electrons and positive ions. If the ions gain enough energy, they will unite to form heavier nuclei, and energy will be released.

Nuclear fusion occurs on the sun and during the explosion of a hydrogen bomb.

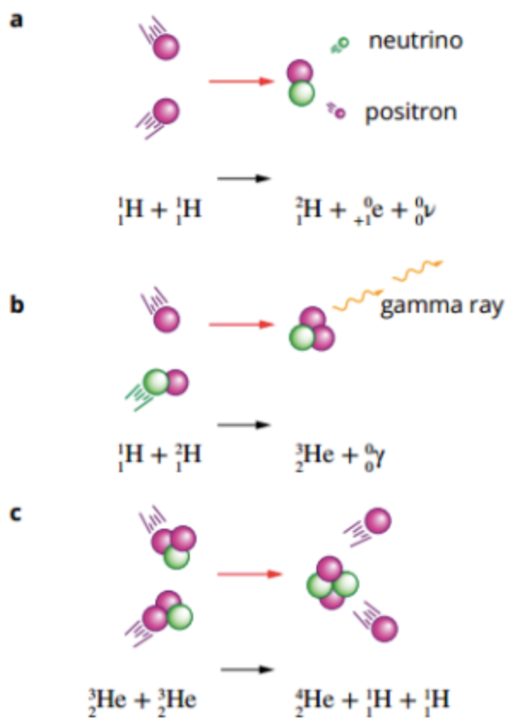


FIGURE 4.3.7 The three main fusion reactions taking place inside in the Sun.

Some typical fusion reactions involving the isotopes of hydrogen (hydrogen $^{1}_{1}H$, deuterium $^{2}_{1}H$ and tritium $^{3}_{1}H$) are as follows.

$$^{2}_{1}H + ^{2}_{1}H \rightarrow ^{3}_{2}He + ^{1}_{0}n + 3.27 \text{ MeV}$$

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + ^{1}_{0}n + 17.6 \text{ MeV}$$

$$^{2}_{1}H + ^{3}_{2}He \rightarrow ^{4}_{2}He + ^{1}_{1}p + 18.3 \text{ MeV}$$

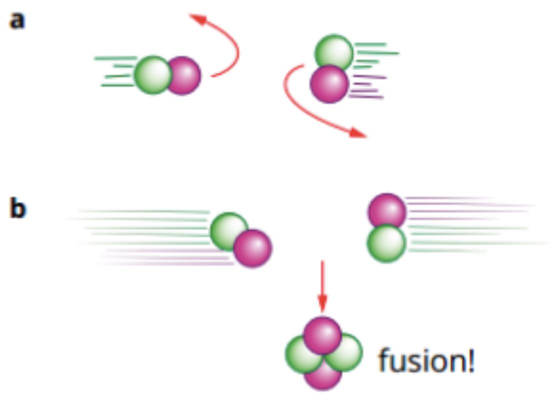


FIGURE 4.3.3 (a) Slow-moving nuclei do not have enough energy to fuse together. The electrostatic forces cause them to be repelled from each other. (b) If the nuclei have sufficient kinetic energy, then they will overcome the repulsive forces and move close enough together for the strong nuclear force to come into effect. At this point, fusion will occur and energy will be released.

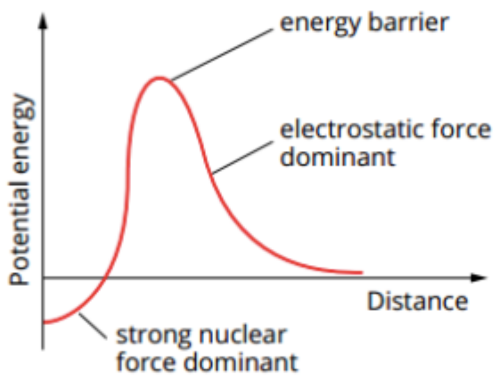


FIGURE 4.3.4 If two hydrogen-2 (deuterium) nuclei are to get close enough for the strong nuclear force to act, they must overcome the energy barrier presented by the electrostatic force.

Fusion Reactors

Several fusion reactions form the helium nucleus, but the one between deuterium and tritium occurs the most easily. This particular reaction holds the greatest promise for achieving a controlled fusion process in the future. The major attractions of fusion power produced from such a process are as follows.

- ***A ready supply of fuel that is almost limitless***. Deuterium is available from seawater while tritium can be produced from beryllium.
- ***Waste products that are non-radioactive***. This is a huge advantage over present fission reactors since it removes the problem of storing radioactive wastes safely for long periods.
- ***Safer reactors***. Fusion reactors dealing with low-density plasma would not be subject to melt down. Reactors could be sited more readily within communities.

The major difficulty in harnessing fusion power is the very high temperatures needed for the reaction. At these high temperatures plasma cannot be easily confined. Research is being undertaken using strong magnetic fields to trap the plasma (very hot ionised gas) that exists during a fusion reaction . At present the most encouraging design is a doughnut shaped device known as a ***tokomak***. It uses large currents to create a toroidal magnetic field that can hold the plasma away from the container walls. Scientists are hopeful that this will lead to a practical fusion reactor in the future.

EXTENSION

Fusion-reactor research

A commercial nuclear fusion reactor is perhaps four or five decades into the future. At the Joint European Torus, or JET, in the UK, research is being carried out into tokamaks—doughnut-shaped reactors that use magnetic fields to contain the plasma of fusion reactants away from the reactor walls and can reach temperatures of hundreds of millions of degrees. The design of the reactor and shape of the torus are shown in figures 4.3.8 and 4.3.9. The largest project in this area is being conducted at the ITER (International Thermonuclear Experimental Reactor). This experiment is a joint venture between Europe, China, India, Japan, Russia, the USA and South Korea, and it is being constructed in France. The aim is to initiate plasma experiments by 2020.

FIGURE 4.3.9 The doughnut shape or torus of the JET fusion reactor can be clearly seen in this photograph.

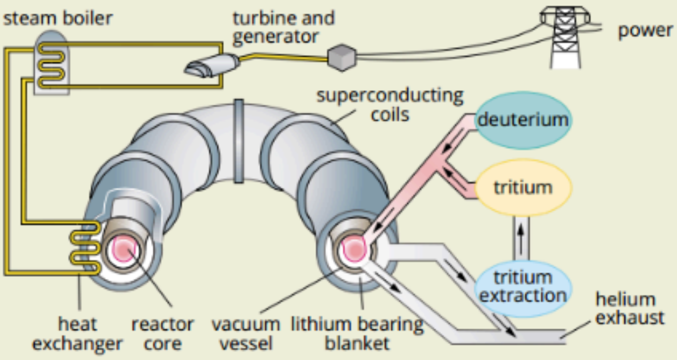


FIGURE 4.3.8 Design of experimental tokamak reactor.

Another technique called inertial confinement is being investigated at the National Ignition Facility in California, USA. Here, a pellet of fusion fuel is zapped by powerful lasers, causing the fuel to implode. This implosion initiates the fusion reaction.

Nuclear fusion may perhaps be the ultimate energy solution. The fuel for nuclear fusion, deuterium, can be readily obtained from seawater. Vast quantities of energy are released during the fusion process, yet a relatively small amount of radioactive waste is created. The radioactive waste consists mostly of the reactor parts that have suffered neutron irradiation.

The Hydrogen Bomb (also known as a thermonuclear bomb)

The temperature produced by a simple fission bomb (atom bomb) can be used to initiate a thermonuclear reaction.

A hydrogen bomb essentially consists of a small atomic bomb surrounded by large amounts of lithium and deuterium. The initial fission reaction raises the temperature sufficiently for fusion to take place between deuterium and tritium. The tritium is formed when neutrons interact with lithium. Vast amounts of energy are released.

PHYSICSFILE

Hydrogen bomb

In 1952, a fusion reaction was used to power the world's first hydrogen bomb. It had five times the destructive power of all the conventional bombs that were dropped during the whole of World War II. The high temperature achieved by a fissile fuel explosion was used to initiate the fusion reaction. In other words, an atomic bomb was used as the fuse for the hydrogen fusion bomb.

FIGURE 4.3.5 The hydrogen bomb dropped at Bikini Atoll in 1956.